Review - The Exponential Model

So far we've considered growth models that look like:

$$\frac{dP}{dt} = kP,$$

for some $k > 0$. This means that the *relative growth rate* is constant:

$$\text{relative growth rate} = \frac{1}{P}\frac{dP}{dt} = k.$$

Can you think of situations where this is not a good model for growth?

Limited Resources - The Logistic Model

When resources are limited, we expect that that relative growth rate gets larger/smaller (cross out the wrong answer) as population grows. Why?

The simplest decreasing function of P is a _____. This can be written as _____.

So we get $\frac{1}{P}\frac{dP}{dt} =$ _____.

For later convenience, we rearrange this as follows:

In general, when there is a limit L to how large $P(t)$ can get, a good model might be:

$$\text{relative growth rate} = \frac{1}{P}\frac{dP}{dt} = k\left(1 - \frac{P}{L}\right).$$

Checking this Makes Sense!

Small Population In an environment with limited resources, if the population is small (near $P = 0$), competition isn't yet a significant factor. Therefore, when P is near 0, we expect the growth to approximate the _____ model.

Check: When $P \approx 0$, then relative growth rate is approximately

$$\underline{\qquad\qquad} = \underline{\quad}.$$

Large Population When $P \approx L$, then relative growth rate is approximately

$$\underline{\qquad\qquad} = \underline{\quad}.$$

Why does this make sense?

What We Expect

Before we dive into the math, let's write down how we expect the population to grow as it increases.

- **Small Population:** When P is small, we expect the population to grow quickly/slowly. Explain.

- **Large Population:** When P is large, we expect the population to grow quickly/slowly. Explain.

- **Midsize Population:** When P is neither too small nor large, we expect the population to grow slowly/quickly. Explain.

Multiplying through by P, we get the **logistic growth model**:

The Logistic Model

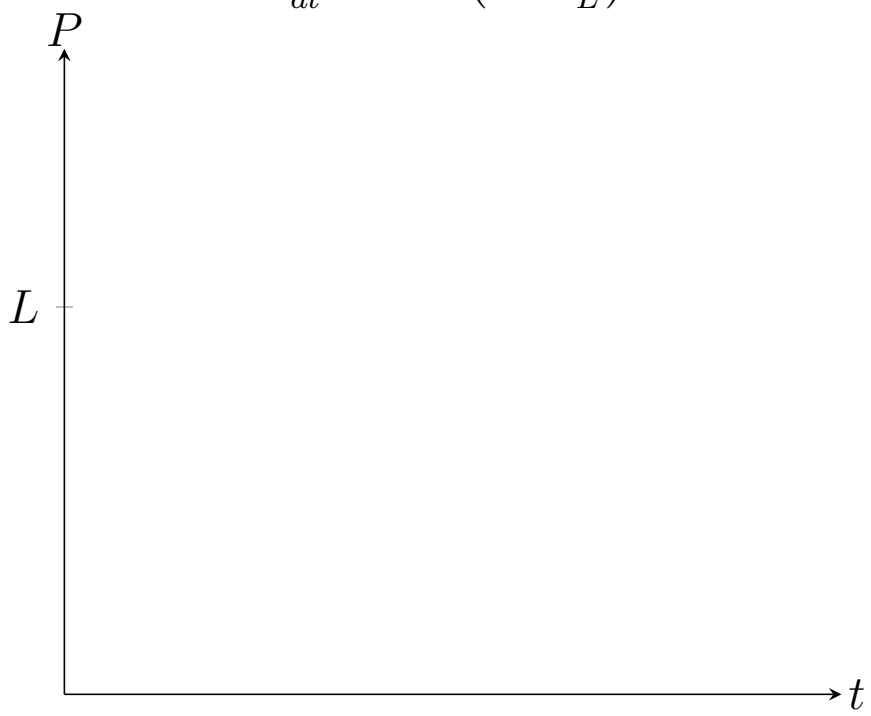
$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$$

Slopefield and Equilibria

Are there any equilibrium solutions?

Use the applet to draw a slopefield for $\frac{dP}{dt}$ and classify any equilibrium solutions as stable or unstable (take $k = 1.5$ and $L = 3$). Sketch two solution curves: one with $P(0)$ near 0 and one with $P(0) > L$. (See the Playing with Differential Equations Lab!)

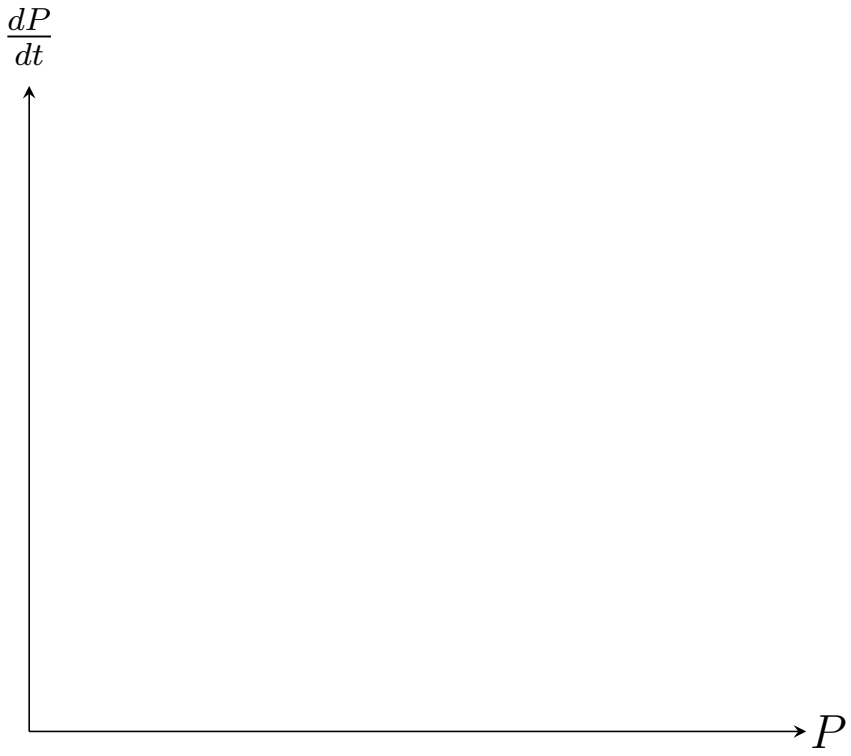
$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$$



Definition As $t \to \infty$, the population approaches L, either from above or below. We call L the *carrying capacity* of the land. It depends on things like the size of the land and availability of resources.

Analyzing the Solutions

Recall from 105L (and Lab 9): when a derivative has an extremum, the function has an _____. Draw a graph of $\frac{dP}{dt}$ as a function of P. For what value of P does $\frac{dP}{dt}$ reach its peak? What is is the maximal value of $\frac{dP}{dt}$? Mark this point on the graph and label its coordinates.



Refer back to your slopefield on the previous page, and the sketch of the solution with $P(0)$ near 0. At the value of P you found in part (a), what feature does the graph of $P(t)$ in that question have? Label this point on the solution. Also draw a tangent line there and label its slope.

Fill in the blanks:

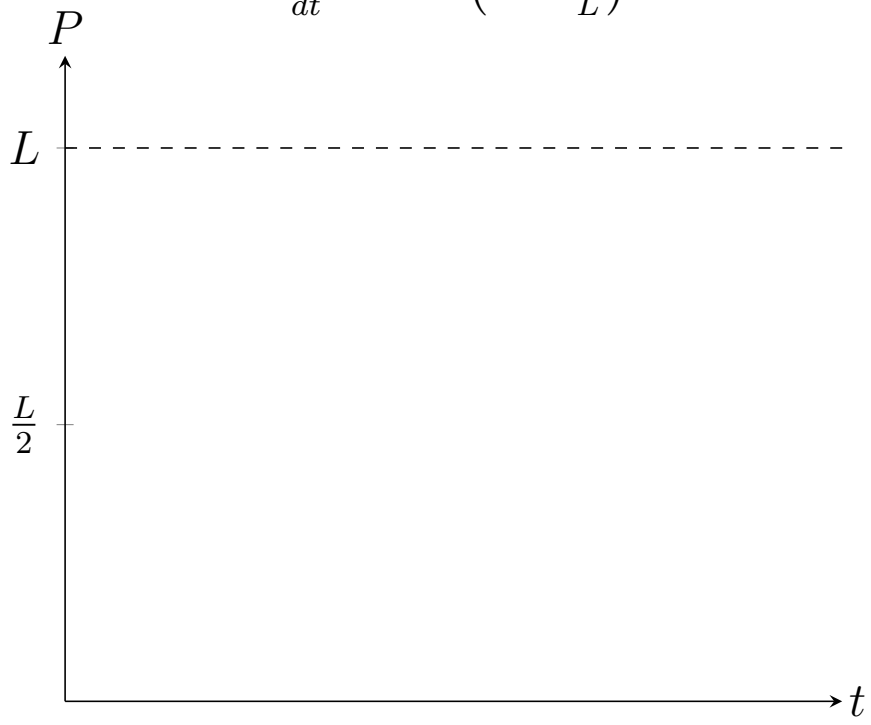
Max Growth Rate

A population that grows logistically has its maximum growth rate when the population is _____ the carrying capacity. The growth rate at that point is _____ organisms per unit time.

Population Growth Above and Below the Max Growth Rate

Draw two solution curves, one with $P(0) < \frac{L}{2}$, and one with $P(0) \geq \frac{L}{2}$ (Use the applet again). What do you observe regarding the concavity of your two solutions?

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$$



The Solution!

Assuming $P(0) = P_0$, the function $P(t) = \frac{L}{1+Ae^{-kt}}$ solves the initial value problem (you can check this by differentiating if you wish). Find A in terms of L and P_0. Show your work.

Assuming that $0 < P_0 < \frac{L}{2}$, how long will it take for $\frac{dP}{dt}$ to reach its maximum value? Show your work.

Fill in the blanks:

Summary of the Logistic Model

The solution to the logistic model:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right),\ P(0) = P_0$$

is the logistic function:

$P(t) =$ _____ with $A =$ _____.

For a logistic function,

The maximum value of $\frac{dP}{dt}$ occurs when $P =$ ____

The maximal value of $\frac{dP}{dt}$ is _____.

Time to the max rate of change: _____ = _____.

Exercise

Suppose that a population grows logistically with $k = 0.1$ years^{-1} and carrying capacity $L = 100$ thousand people. Suppose also that the initial population under consideration is 5,000 people. You may use any parts of the summary above to do the following problem.

1. Write down an initial value problem for this situation, and its solution.

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2. (a) How many people are on the land when the growth rate of the population is maximal?

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(b) How long does it take for the population to reach its maximal growth rate? Include units.

(c) What is the maximal growth rate of the population? Express your answer in people/year.

3. How long does it take for the population to reach 95% of the carrying capacity of the land? What is the growth rate (in people/year) at that point in time? Include units.

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4. Sketch a graph of the population over time. Mark both of the points from the previous two questions on your graph. Also draw tangent lines to your graph at those points and label their slopes.

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